Step 1: Dataset

1.1 Import Dataset

In [1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("311_Service_Requests_from_2010_to_Present.csv", low_memory=Fal
df.head()
```

Out[1]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor
0	32310363	12/31/2015 11:59:45 PM	01/01/2016 12:55:15 AM	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party
1	32309934	12/31/2015 11:59:44 PM	01/01/2016 01:26:57 AM	NYPD	New York City Police Department	Blocked Driveway	No Access
2	32309159	12/31/2015 11:59:29 PM	01/01/2016 04:51:03 AM	NYPD	New York City Police Department	Blocked Driveway	No Access
3	32305098	12/31/2015 11:57:46 PM	01/01/2016 07:43:13 AM	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking
4	32306529	12/31/2015 11:56:58 PM	01/01/2016 03:24:42 AM	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk

5 rows × 53 columns

1.2 Visualize Dataset

In [2]:
```
df.head(10)
```

Out[2]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor
0	32310363	12/31/2015 11:59:45 PM	01/01/2016 12:55:15 AM	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party
1	32309934	12/31/2015 11:59:44 PM	01/01/2016 01:26:57 AM	NYPD	New York City Police Department	Blocked Driveway	No Access
2	32309159	12/31/2015 11:59:29 PM	01/01/2016 04:51:03 AM	NYPD	New York City Police Department	Blocked Driveway	No Access
3	32305098	12/31/2015 11:57:46 PM	01/01/2016 07:43:13 AM	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking
4	32306529	12/31/2015 11:56:58 PM	01/01/2016 03:24:42 AM	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk
5	32306554	12/31/2015 11:56:30 PM	01/01/2016 01:50:11 AM	NYPD	New York City Police Department	Illegal Parking	Posted Parking Sign Violation
6	32306559	12/31/2015 11:55:32 PM	01/01/2016 01:53:54 AM	NYPD	New York City Police Department	Illegal Parking	Blocked Hydrant
7	32307009	12/31/2015 11:54:05 PM	01/01/2016 01:42:54 AM	NYPD	New York City Police Department	Blocked Driveway	No Access
8	32308581	12/31/2015 11:53:58 PM	01/01/2016 08:27:32 AM	NYPD	New York City Police Department	Illegal Parking	Posted Parking Sign Violation
9	32308391	12/31/2015 11:53:58 PM	01/01/2016 01:17:40 AM	NYPD	New York City Police Department	Blocked Driveway	No Access

10 rows × 53 columns

1.3 Columns

In [3]:
```
print(df.columns)
```

```
Index(['Unique Key', 'Created Date', 'Closed Date', 'Agency', 'Agency Name',
       'Complaint Type', 'Descriptor', 'Location Type', 'Incident Zip',
       'Incident Address', 'Street Name', 'Cross Street 1', 'Cross Street 2',
       'Intersection Street 1', 'Intersection Street 2', 'Address Type',
       'City', 'Landmark', 'Facility Type', 'Status', 'Due Date',
       'Resolution Description', 'Resolution Action Updated Date',
       'Community Board', 'Borough', 'X Coordinate (State Plane)',
       'Y Coordinate (State Plane)', 'Park Facility Name', 'Park Borough',
       'School Name', 'School Number', 'School Region', 'School Code',
       'School Phone Number', 'School Address', 'School City', 'School State',
       'School Zip', 'School Not Found', 'School or Citywide Complaint',
       'Vehicle Type', 'Taxi Company Borough', 'Taxi Pick Up Location',
       'Bridge Highway Name', 'Bridge Highway Direction', 'Road Ramp',
       'Bridge Highway Segment', 'Garage Lot Name', 'Ferry Direction',
       'Ferry Terminal Name', 'Latitude', 'Longitude', 'Location'],
      dtype='object')
```

1.4 Shape

In [4]:
```
print("Shape:", df.shape)
```

```
Shape: (364558, 53)
```

1.5 Null Values

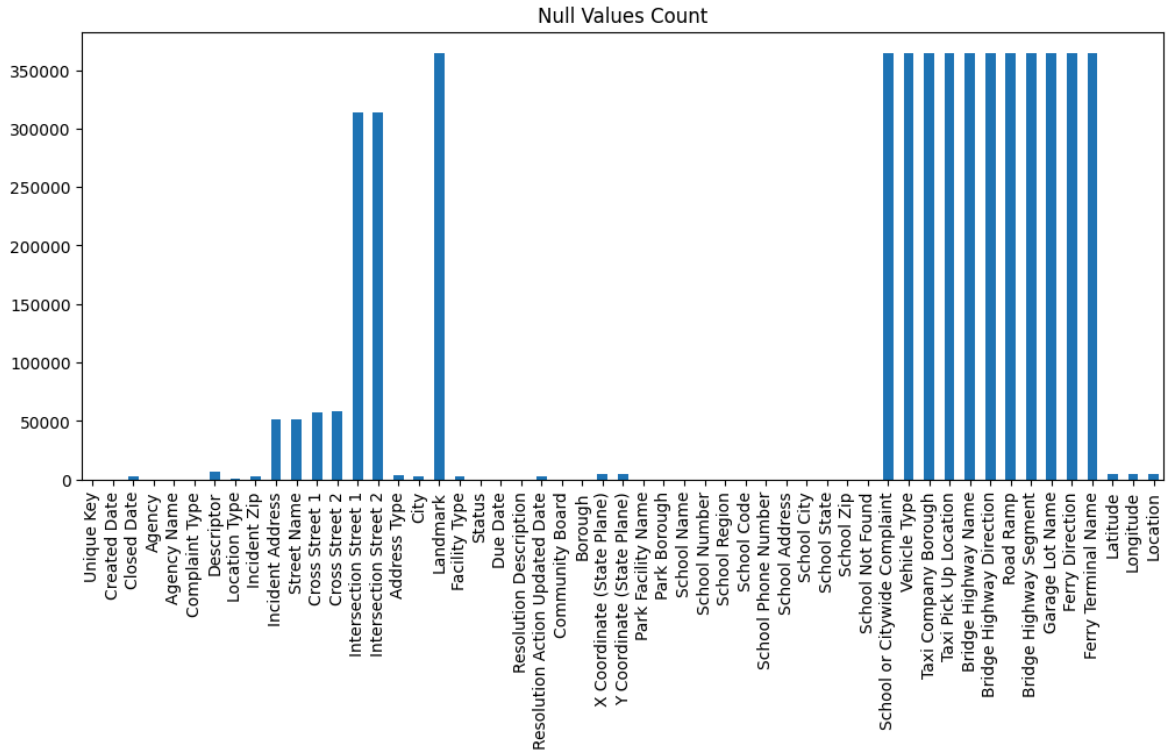
In [5]:
```
df.isnull().sum()
```

Out[5]:
```
Unique Key                          0
Created Date                        0
Closed Date                      2381
Agency                              0
Agency Name                         0
Complaint Type                      0
Descriptor                       6501
Location Type                     133
Incident Zip                     2998
Incident Address                51699
Street Name                     51699
Cross Street 1                  57188
Cross Street 2                  57805
Intersection Street 1          313438
Intersection Street 2          314046
Address Type                     3252
City                             2997
Landmark                       364183
Facility Type                    2389
Status                              0
Due Date                            3
Resolution Description              0
Resolution Action Updated Date   2402
Community Board                     0
Borough                             0
X Coordinate (State Plane)       4030
Y Coordinate (State Plane)       4030
Park Facility Name                  0
Park Borough                        0
School Name                         0
School Number                       0
School Region                       1
School Code                         1
School Phone Number                 0
School Address                      0
School City                         0
School State                        0
School Zip                          1
School Not Found                    0
School or Citywide Complaint   364558
Vehicle Type                   364558
Taxi Company Borough           364558
Taxi Pick Up Location          364558
Bridge Highway Name            364261
Bridge Highway Direction       364261
Road Ramp                      364296
Bridge Highway Segment         364296
Garage Lot Name                364558
Ferry Direction                364557
Ferry Terminal Name            364556
Latitude                         4030
Longitude                        4030
Location                         4030
dtype: int64
```

Step 2: Data Analysis

2.1 Null Values Bar Plot

In [6]:
```
df.isnull().sum().plot(kind='bar', figsize=(12,5))
plt.title("Null Values Count")
plt.show()
```



2.2 Missing Value Treatment

Remove Closed Date null

In [13]:
```
df = df[df['Closed Date'].notnull()]
```

In [14]:
```
print("After removing null values:", df.shape)
df.head()
```

```
After removing null values: (362177, 54)
```

Out[14]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor	Locat
0	32310363	2015-12-31 23:59:45	2016-01-01 00:55:15	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/
1	32309934	2015-12-31 23:59:44	2016-01-01 01:26:57	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/
2	32309159	2015-12-31 23:59:29	2016-01-01 04:51:03	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/
3	32305098	2015-12-31 23:57:46	2016-01-01 07:43:13	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking	Street/
4	32306529	2015-12-31 23:56:58	2016-01-01 03:24:42	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk	Street/

5 rows × 54 columns

2.3 Date Analysis

In [9]:
```
df['Created Date'] = pd.to_datetime(df['Created Date'], format='%m/%d/%Y %I:%M:%
df['Closed Date'] = pd.to_datetime(df['Closed Date'], format='%m/%d/%Y %I:%M:%S
```

In [15]:
```
df[['Created Date','Closed Date']].head()
df.dtypes
```

Out[15]:
```
Unique Key                              int64
Created Date                     datetime64[ns]
Closed Date                      datetime64[ns]
Agency                                  object
Agency Name                             object
Complaint Type                          object
Descriptor                              object
Location Type                           object
Incident Zip                           float64
Incident Address                        object
Street Name                             object
Cross Street 1                          object
Cross Street 2                          object
Intersection Street 1                   object
Intersection Street 2                   object
Address Type                            object
City                                    object
Landmark                                object
Facility Type                           object
Status                                  object
Due Date                                object
Resolution Description                  object
Resolution Action Updated Date          object
Community Board                         object
Borough                                 object
X Coordinate (State Plane)             float64
Y Coordinate (State Plane)             float64
Park Facility Name                      object
Park Borough                            object
School Name                             object
School Number                           object
School Region                           object
School Code                             object
School Phone Number                     object
School Address                          object
School City                             object
School State                            object
School Zip                              object
School Not Found                        object
School or Citywide Complaint           float64
Vehicle Type                           float64
Taxi Company Borough                   float64
Taxi Pick Up Location                  float64
Bridge Highway Name                     object
Bridge Highway Direction                object
Road Ramp                               object
Bridge Highway Segment                  object
Garage Lot Name                        float64
Ferry Direction                         object
Ferry Terminal Name                     object
Latitude                               float64
Longitude                              float64
Location                                object
Request_Closing_Time                   float64
dtype: object
```

2.3.1 Time Difference

In [10]:
```
df['Request_Closing_Time'] = df['Closed Date'] - df['Created Date']
```

In [16]:
```
df[['Created Date','Closed Date','Request_Closing_Time']].head()
```

Out[16]:

	Created Date	Closed Date	Request_Closing_Time
0	2015-12-31 23:59:45	2016-01-01 00:55:15	3330.0
1	2015-12-31 23:59:44	2016-01-01 01:26:57	5233.0
2	2015-12-31 23:59:29	2016-01-01 04:51:03	17494.0
3	2015-12-31 23:57:46	2016-01-01 07:43:13	27927.0
4	2015-12-31 23:56:58	2016-01-01 03:24:42	12464.0

2.3.2 Convert to seconds

In [11]:
```
df['Request_Closing_Time'] = df['Request_Closing_Time'].dt.total_seconds()
```

In [17]:
```
df[['Request_Closing_Time']].head()
```

Out[17]:

	Request_Closing_Time
0	3330.0
1	5233.0
2	17494.0
3	27927.0
4	12464.0

2.3.3 Statistics

In [12]:
```
df['Request_Closing_Time'].describe()
```

Out[12]:
```
count    3.621770e+05
mean     1.511330e+04
std      2.110255e+04
min      6.100000e+01
25%      4.533000e+03
50%      9.616000e+03
75%      1.887800e+04
max      2.134342e+06
Name: Request_Closing_Time, dtype: float64
```

2.3.4 Null Check

In [18]:
```
df[['Complaint Type','City']].isnull().sum()
```

Out[18]:
```
Complaint Type      0
City              674
dtype: int64
```

2.3.5 Fill Missing City

In [19]:
```
df['City'].fillna("Unknown City", inplace=True)
```

In [20]:
```
print(df.shape)
df.head()
```

```
(362177, 54)
```

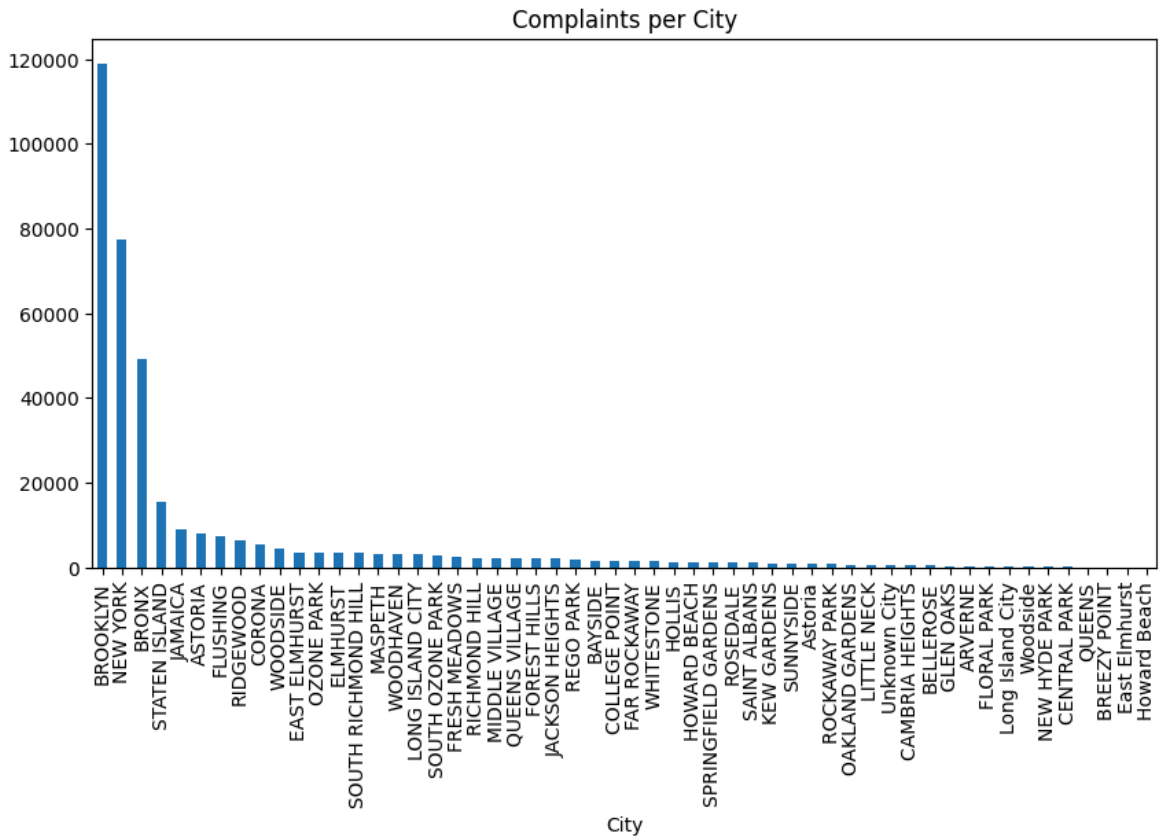
Out[20]:

	Unique Key	Created Date	Closed Date	Agency	Agency Name	Complaint Type	Descriptor	Locat
0	32310363	2015-12-31 23:59:45	2016-01-01 00:55:15	NYPD	New York City Police Department	Noise - Street/Sidewalk	Loud Music/Party	Street/
1	32309934	2015-12-31 23:59:44	2016-01-01 01:26:57	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/
2	32309159	2015-12-31 23:59:29	2016-01-01 04:51:03	NYPD	New York City Police Department	Blocked Driveway	No Access	Street/
3	32305098	2015-12-31 23:57:46	2016-01-01 07:43:13	NYPD	New York City Police Department	Illegal Parking	Commercial Overnight Parking	Street/
4	32306529	2015-12-31 23:56:58	2016-01-01 03:24:42	NYPD	New York City Police Department	Illegal Parking	Blocked Sidewalk	Street/

5 rows × 54 columns

2.3.6 Complaints per City

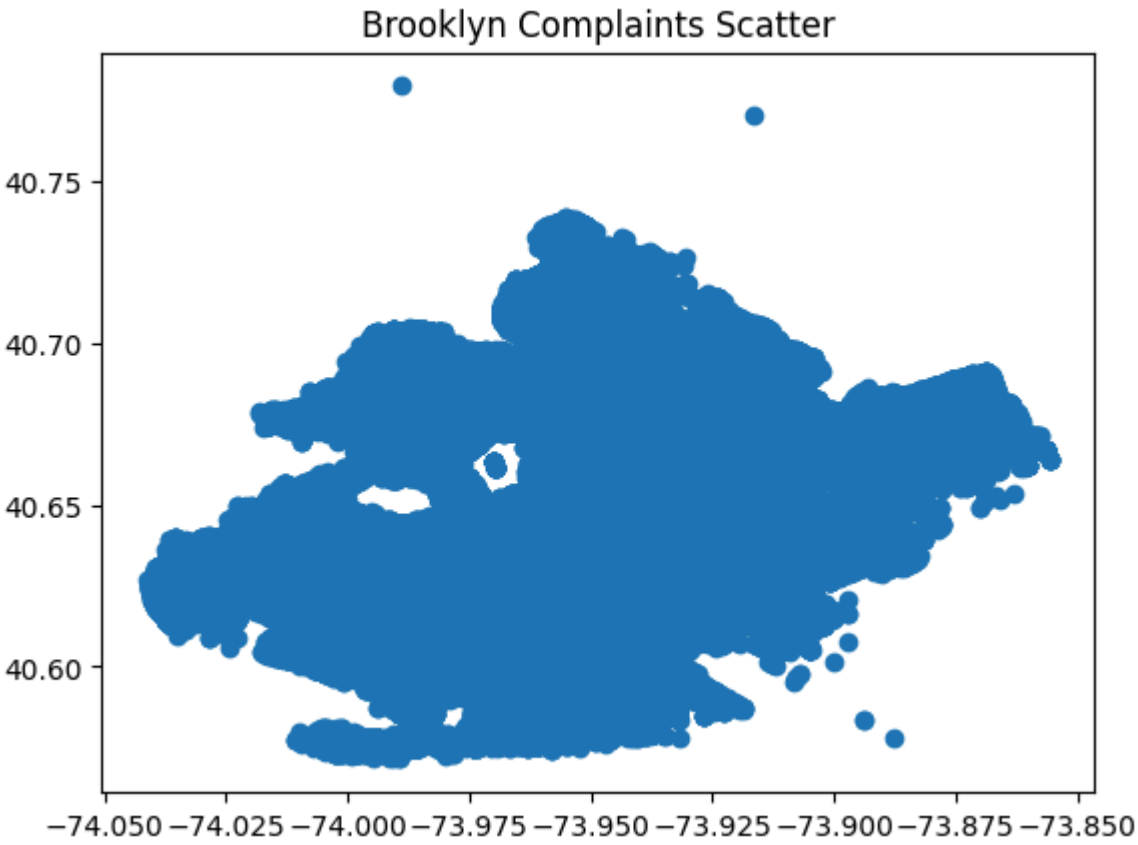
In [21]:
```
df['City'].value_counts().plot(kind='bar', figsize=(10,5))
plt.title("Complaints per City")
plt.show()
```



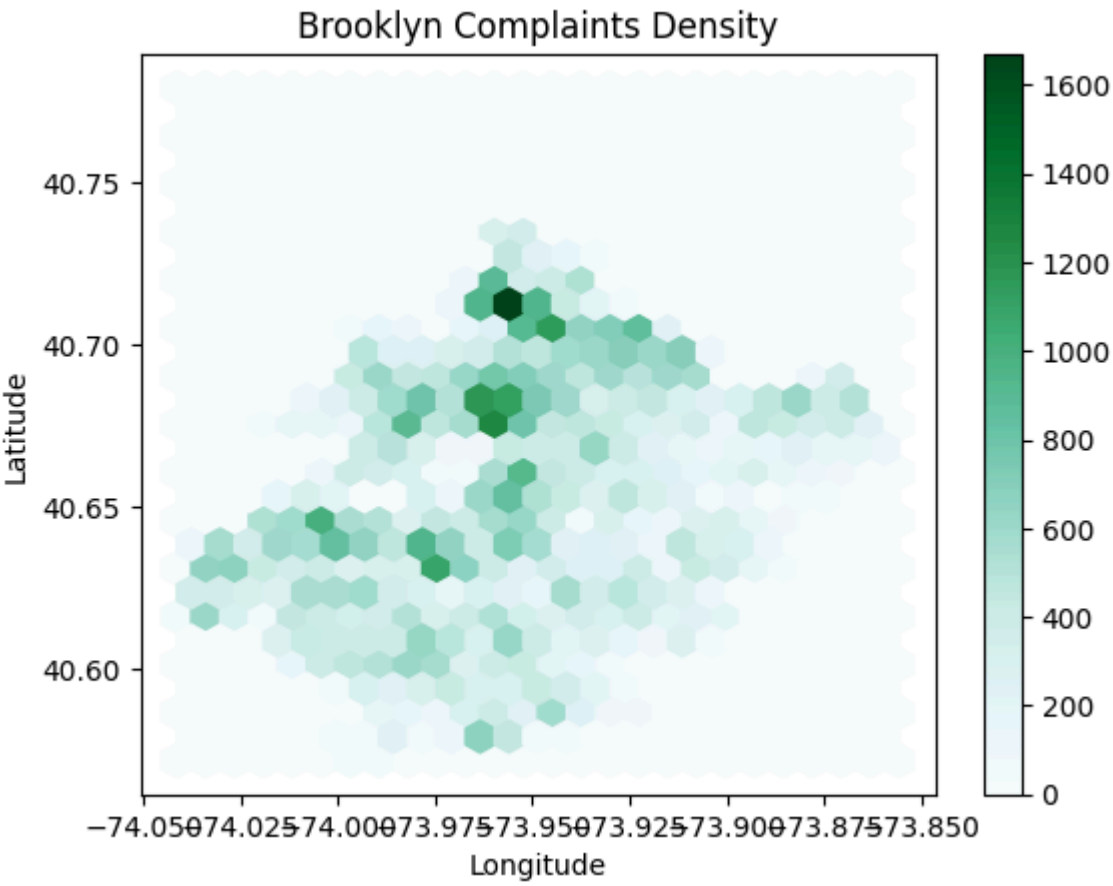
2.3.7 Brooklyn Scatter + Hexbin

In [22]:
```
brooklyn = df[df['City'] == 'BROOKLYN']

plt.scatter(brooklyn['Longitude'], brooklyn['Latitude'])
plt.title("Brooklyn Complaints Scatter")
plt.show()
```



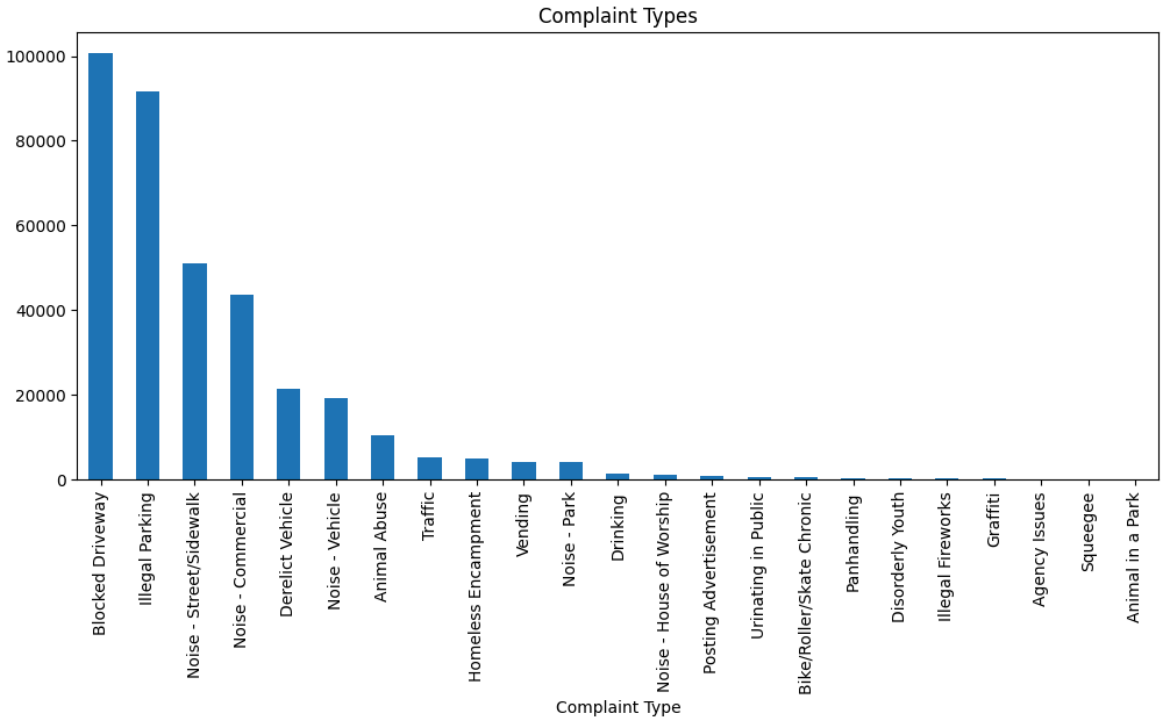
In [23]:
```
brooklyn.plot.hexbin(x='Longitude', y='Latitude', gridsize=25)
plt.title("Brooklyn Complaints Density")
plt.show()
```



Step 3: Major Complaint Types

3.1 Complaint Types Bar

In [24]:
```
df['Complaint Type'].value_counts().plot(kind='bar', figsize=(12,5))
plt.title("Complaint Types")
plt.show()
```



3.2 NYC Complaints

In [25]:
```
nyc = df[df['City'] == 'NEW YORK']
nyc['Complaint Type'].value_counts()
```

Out[25]:
```
Complaint Type
Noise - Street/Sidewalk      22245
Noise - Commercial           18686
Illegal Parking              14549
Noise - Vehicle               6294
Homeless Encampment           3060
Blocked Driveway              2705
Vending                       2638
Animal Abuse                  1941
Traffic                       1769
Noise - Park                  1243
Derelict Vehicle               695
Drinking                       321
Urinating in Public            264
Bike/Roller/Skate Chronic      254
Noise - House of Worship       222
Panhandling                    206
Disorderly Youth                81
Posting Advertisement           49
Illegal Fireworks               38
Graffiti                        25
Squeegee                         4
Name: count, dtype: int64
```

3.3 Top 10 Complaints

In [26]:
```
df['Complaint Type'].value_counts().head(10)
```

Out[26]:
```
Complaint Type
Blocked Driveway           100624
Illegal Parking             91716
Noise - Street/Sidewalk     51139
Noise - Commercial          43751
Derelict Vehicle            21518
Noise - Vehicle             19301
Animal Abuse                10530
Traffic                      5196
Homeless Encampment          4879
Vending                      4185
Name: count, dtype: int64
```

3.4 Complaints per City

In [27]:
```
pd.crosstab(df['City'], df['Complaint Type'])
```

Out[27]:

Complaint Type City	Agency Issues	Animal Abuse	Animal in a Park	Bike/Roller/Skate Chronic	Blocked Driveway	Derelict Vehicle	Disorder You
ARVERNE	0	46	0	0	50	32	
ASTORIA	0	170	0	16	3436	426	
Astoria	0	0	0	0	159	14	
BAYSIDE	0	53	0	0	514	231	
BELLEROSE	0	15	0	1	138	120	
BREEZY POINT	0	2	0	0	3	3	
BRONX	0	1971	0	22	17062	2402	
BROOKLYN	0	3191	0	124	36445	6257	
CAMBRIA HEIGHTS	0	15	0	0	177	148	
CENTRAL PARK	0	0	0	0	0	0	
COLLEGE POINT	0	35	0	0	597	223	
CORONA	0	104	0	0	3597	72	
EAST ELMHURST	0	85	0	1	1925	136	
ELMHURST	0	59	0	2	1992	94	
East Elmhurst	0	0	0	0	0	2	
FAR ROCKAWAY	0	111	0	0	383	215	
FLORAL PARK	0	7	0	0	33	74	
FLUSHING	0	191	0	3	3640	532	
FOREST HILLS	0	78	0	6	873	71	
FRESH MEADOWS	0	66	0	0	682	347	
GLEN OAKS	0	5	0	0	48	57	
HOLLIS	0	39	0	0	442	162	
HOWARD BEACH	0	51	0	1	215	172	
Howard Beach	0	0	0	0	1	0	

Complaint Type City	Agency Issues	Animal Abuse	Animal in a Park	Bike/Roller/Skate Chronic	Blocked Driveway	Derelict Vehicle	Disorder You
JACKSON HEIGHTS	0	50	0	2	703	41	
JAMAICA	0	317	0	3	3620	1132	
KEW GARDENS	0	26	0	0	429	16	
LITTLE NECK	0	21	0	0	174	73	
LONG ISLAND CITY	0	40	0	3	1052	220	
Long Island City	0	0	0	0	55	4	
MASPETH	0	56	0	1	1000	510	
MIDDLE VILLAGE	0	36	0	1	663	366	
NEW HYDE PARK	0	1	0	0	76	14	
NEW YORK	0	1941	0	254	2705	695	
OAKLAND GARDENS	0	29	0	2	177	117	
OZONE PARK	0	72	0	1	1681	479	
QUEENS	0	1	1	0	3	2	
QUEENS VILLAGE	0	90	0	0	772	478	
REGO PARK	0	33	0	0	780	94	
RICHMOND HILL	0	55	0	0	1099	200	
RIDGEWOOD	0	154	0	3	2161	507	
ROCKAWAY PARK	0	33	0	0	80	19	
ROSEDALE	0	44	0	2	270	247	
SAINT ALBANS	0	43	0	0	318	248	
SOUTH OZONE PARK	0	74	0	1	1202	425	
SOUTH RICHMOND HILL	0	40	0	1	1946	356	

Complaint Type	Agency Issues	Animal Abuse	Animal in a Park	Bike/Roller/Skate Chronic	Blocked Driveway	Derelict Vehicle	Disorde You
City							
SPRINGFIELD GARDENS	0	42	0	0	330	267	
STATEN ISLAND	0	786	0	10	2845	2184	
SUNNYSIDE	0	40	0	2	278	17	
Unknown City	8	1	0	2	86	63	
WHITESTONE	0	43	0	4	279	279	
WOODHAVEN	0	57	0	2	1363	369	
WOODSIDE	0	111	0	5	2038	298	
Woodside	0	0	0	0	27	8	

54 rows × 23 columns

3.5 New DataFrame

In [28]:
```
df_new = pd.crosstab(df['Complaint Type'], df['City'])
df_new.head()
```

Out[28]:

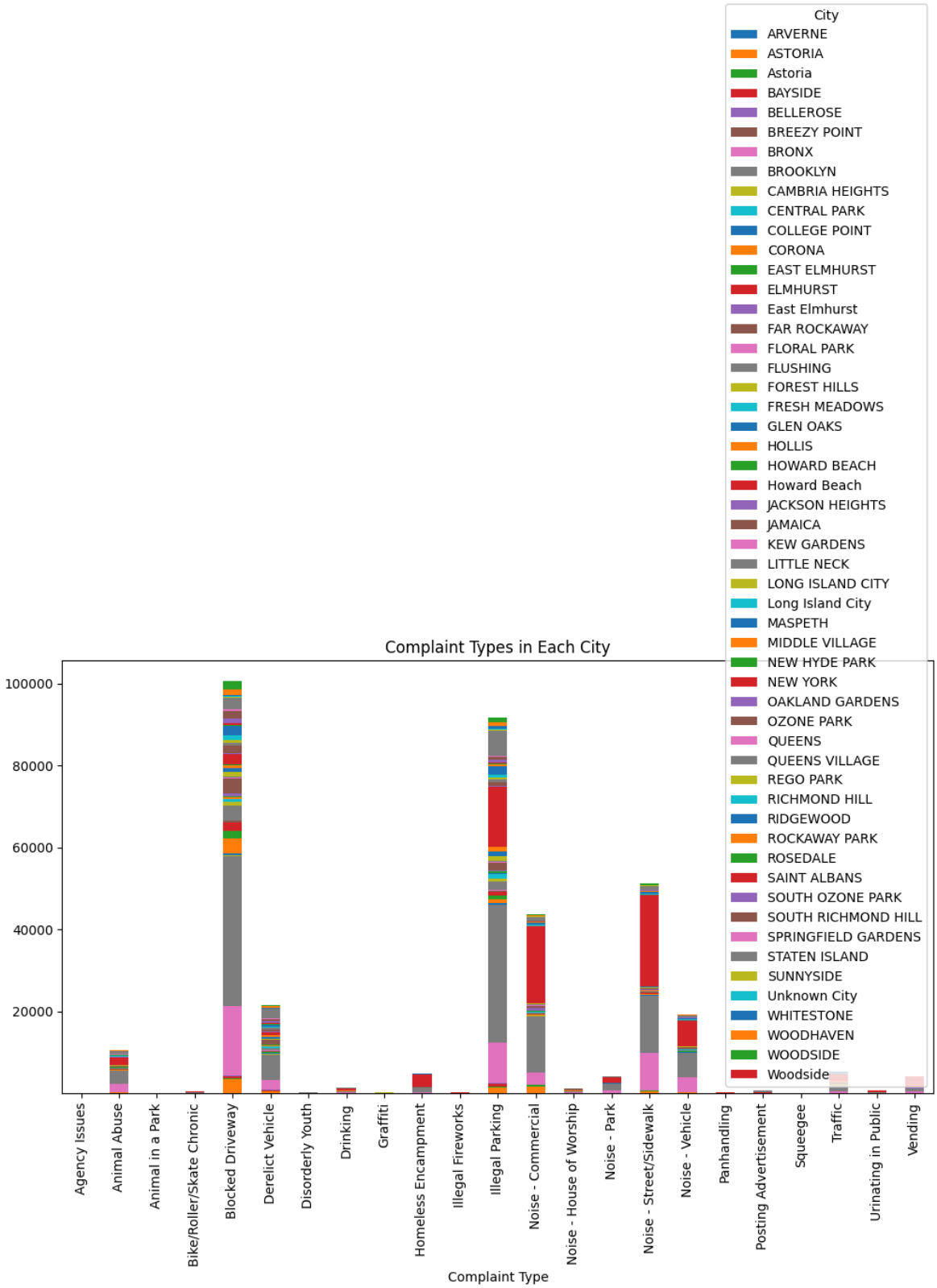
City	ARVERNE	ASTORIA	Astoria	BAYSIDE	BELLEROSE	BREEZY POINT	BRONX
Complaint Type							
Agency Issues	0	0	0	0	0	0	0
Animal Abuse	46	170	0	53	15	2	1971
Animal in a Park	0	0	0	0	0	0	0
Bike/Roller/Skate Chronic	0	16	0	0	1	0	22
Blocked Driveway	50	3436	159	514	138	3	17062

5 rows × 54 columns

Step 4: Visualization

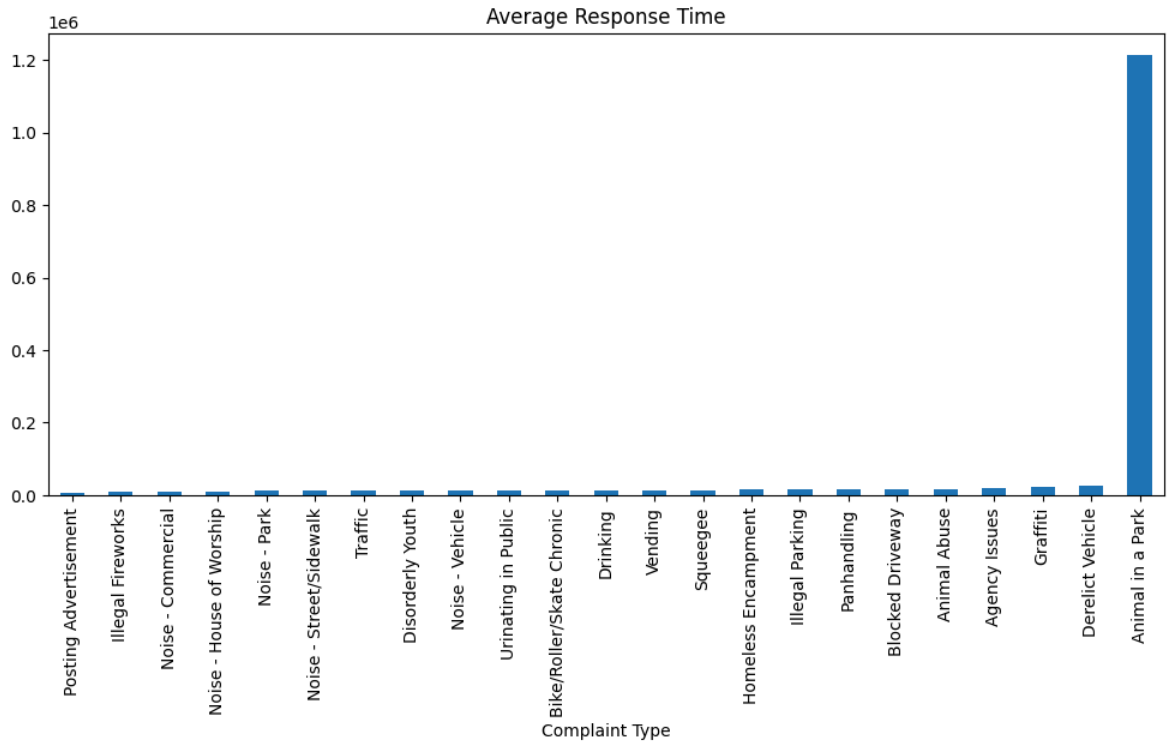
4.1 Combined Chart

In [32]:
```
df_new.plot(kind='bar', stacked=True, figsize=(12,6))
plt.title("Complaint Types in Each City")
plt.show()
```



4.2 Average Closing Time

In [33]:
```
avg_time = df.groupby('Complaint Type')['Request_Closing_Time'].mean()
avg_time.sort_values().plot(kind='bar', figsize=(12,5))
plt.title("Average Response Time")
plt.show()
```



Step 5: Response Time Analysis

In [34]:
```
df.groupby('Complaint Type')['Request_Closing_Time'].mean()
```

Out[34]:
```
Complaint Type
Agency Issues                1.828912e+04
Animal Abuse                 1.803256e+04
Animal in a Park             1.212634e+06
Bike/Roller/Skate Chronic    1.312369e+04
Blocked Driveway             1.623252e+04
Derelict Vehicle             2.535960e+04
Disorderly Youth             1.236375e+04
Drinking                     1.382130e+04
Graffiti                     2.327634e+04
Homeless Encampment          1.545138e+04
Illegal Fireworks            1.011348e+04
Illegal Parking              1.565044e+04
Noise - Commercial           1.108576e+04
Noise - House of Worship     1.139109e+04
Noise - Park                 1.222606e+04
Noise - Street/Sidewalk      1.223130e+04
Noise - Vehicle              1.256180e+04
Panhandling                  1.585355e+04
Posting Advertisement        7.286256e+03
Squeegee                     1.456025e+04
Traffic                      1.230912e+04
Urinating in Public          1.295929e+04
Vending                      1.436628e+04
Name: Request_Closing_Time, dtype: float64
```

Step 6 & 7: Statistical Test (Kruskal-Wallis)

In [35]:
```
from scipy.stats import kruskal

groups = []
```

```
for name, group in df.groupby('Complaint Type'):
    groups.append(group['Request_Closing_Time'].dropna())

stat, p = kruskal(*groups)

print("P-value:", p)

if p < 0.05:
    print("Reject H0 (Different response times)")
else:
    print("Fail to Reject H0 (Same response times)")
```

```
P-value: 0.0
Reject H0 (Different response times)
```

Step 8: Observations

Most complaints are concentrated in specific cities like Brooklyn and New York.

• Major complaint types include Noise, Illegal Parking, and Blocked Driveway.

• Response time varies significantly across complaint types.

• Some complaints are resolved faster than others.

• Data cleaning improved the quality of analysis.

• Statistical test indicates whether response time differences are significant.